BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, GUJRANWALA

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

CHEMISTRY

Past Paper 2024 • Paper Code: 8483 / 8488

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Chemistry (Intermediate Part-II) Paper : II (Objective Type)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8483

Note: Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

Q1. 1. The main pollutant of leather tanneries in the waste water is due to salt of

(A) Lead
(B) Chromium (VI)
(C) Copper
(D) Chromium (III)

Q2. One of the following will have the highest boiling point

(A) methanal
(B) ethanal
(C) propanal
(D) 2 – hexanone

Q3. The normal amount of overhead Ozone is about

(A) 150 DU
(B) 250 DU
(C) 350 DU
(D) 450 DU

Q4. Ethanol can be converted into ethanoic acid by

(A) Hydrogenation
(B) Hydration
(C) Oxidation
(D) Fermentation

Q5. Phosphorus helps the growth of

(A) root
(B) leaf
(C) stem
(D) seed

Q6. When ethanal ($CH_3-C(=O)-H$) is made to react with ethyl Magnesium Bromide followed by acid hydrolysis, the product formed is

(A) 1–propanol
(B) 2–propanol
(C) 1–butanol
(D) 2–butanol

Q7. Which one of the following elements is not present in all proteins?

(A) Sulphur
(B) Carbon
(C) Hydrogen
(D) Nitrogen

Q8. The anhydride of $HClO_4$ is

(A) ClO_3
(B) ClO_2
(C) Cl_2O_5
(D) Cl_2O_7

Q9. Which reagent is used to reduce a Carboxylic group to an alcohol?

(A) H_2/Ni
(B) H_2/Pt
(C) $NaBH_4$
(D) $LiAlH_4$

Q10. In ring test, the colour of $FeSO_4 \cdot NO$ is

(A) Brown
(B) Red
(C) Green
(D) Black

Q11. The conversion of n-hexane into benzene by heating in the presence of Pt is called

(A) Isomerization
(B) Aromatization
(C) de-alkylation
(D) Re-arrangement

Q12. The chief ore of aluminium is

(A) Na_3AlF_6
(B) $Al_2O_3 \cdot 2H_2O$
(C) Al_2O_3
(D) $Al_2O_3 \cdot H_2O$

Q13. β–β′–dichloroethyl sulphide is commonly known as

(A) Laughing gas
(B) Bio-gas
(C) Mustard gas
(D) Phosgene gas

Q14. One of the following is not an alkali metal

(A) Francium
(B) Caesium
(C) Rubidium
(D) Radium

Q15. The state of hybridization of Carbon atom in Methane is

(A) sp^3
(B) sp^2
(C) sp
(D) dsp^2

Q16. Select the two elements which are present in third period

(A) Li , Be
(B) Na , Mg
(C) K , Ca
(D) Rb , Sr

Q17. One of the following is a typical transition metal

(A) Sc
(B) Y
(C) Co
(D) Ra

Chemistry (Intermediate Part-II) Paper : II (Subjective Type)

Time Allowed: 2.40 hours | **Maximum Marks:** 68

Note: Section I is compulsory. Attempt any THREE (3) questions from Section II.

Q2. (Marks (2 x 8 = 16)) Write short answers to any EIGHT questions.

i. Why the ionic radii of negative ions are larger than the size of their parent atoms? Give example.

ii. Give two defects in Mendeleev's periodic table.

iii. Why transition elements have variable oxidation states?

iv. $KMnO_4$ acts as oxidizing agent? Justify with two examples.

v. Write the chemistry of setting of cement in first twenty four hours.

vi. Alkali metals give ionic hydrides. Give reason.

vii. Write the formula of (a) Asbestos (b) Halite

viii. What is the excellent method to prepare Alkyl Iodide?

ix. What is terpolymer? Give one example.

x. What are the monosaccharides? Give one example.

xi. Compare the compound protein with derived protein.

xii. How Grignard reagent reacts with "HCHO"?

Q3. (Marks (2 x 8 = 16)) Write short answers to any EIGHT questions.

i. What is aqua regia? How does it dissolve gold?

ii. NO_2 is oxidizing agent. Prove it with two suitable examples.

iii. Arrange the oxidizing power of following with reason: F_2 , Cl_2 , Br_2 , I_2

iv. Why I_2 is solid while F_2 is gas?

v. Define functional group with any two examples.

vi. Differentiate between metamerism and position isomerism, with suitable examples.

vii. How oxalic acid is prepared from acetylene?

viii. Why ethyne is less reactive than ethene?

ix. Why alkanes are called as paraffins?

x. What are leachate?

xi. Define Biochemical Oxygen demand.

xii. What is lithosphere?

Q4. (Marks (2 x 6 = 12)) Write short answers to any SIX questions.

i. How to prepare borax from coleminte?

ii. Give two important uses of silicates.

iii. What is white lead? Give its use.

iv. Write IUPAC names of the following molecules: a) [structure: benzene ring with one CH_3 group and three NO_2 groups] b) [structure: benzene ring with one OH group and one CH_3 group at para position]

v. Ethyl alcohol is a liquid while methyl chloride is a gas, justify it.

vi. Give the reaction of phenol with: (a) Zn (b) HNO_3

vii. How acetaldehyde reacts with the following reagents? (a) HCN (b) I_2/NaOH

viii. What is vinegar and give its use?

ix. Write down the mechanism of reaction between acetic acid and methanol.

> Note: SECTION – II

Q5.

(a) [1+1+2=4] What is hydration energy? Give one example. Discuss its variation in groups and periods of periodic table.

(b) [4] Write down any eight uses of lime in industry.

Q6.

(a) [4] Give the reactions of Bleaching powder with i. dil H_2SO_4 ii. HCl iii. NH_3 iv. CO_2

(b) [4] Describe the process of digestion of paper pulp in Neutral Sulphite chemical process.

Q7.

(a) [4] What are organic compounds? Describe the following terms (give one example for each) i. Alicyclic Compounds ii. Aromatic Compounds iii. Heterocyclic Compounds

(b) [4] Write down the chemical reaction of Ethyl Magnesium Bromide with CO_2 , HCHO , Acetone and Epoxide.

Q8.

(a) [4] Explain acidic nature of alkyne in detail by giving two examples.

(b) [4] What is aldol condensation? Give its mechanism.

Q9.

(a) [4] Discuss catalytic oxidation of Benzene.

(b) [4] How ethers are prepared from following? i. Williamsons Synthesis ii. Ag_2O

GROUP – II

Chemistry (Intermediate Part-II) Paper : II (Objective Type)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8488

Note: Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

Q1. 1. Which one of the following compounds is most reactive?

(A) Benzene
(B) Ethene
(C) Ethane
(D) Ethyne

Q2. Which one of the following is a non-typical transition element?

(A) Cr
(B) Mn
(C) Zn
(D) Fe

Q3. β–β′–dichloroethyl sulphide is commonly known as

(A) Laughing gas
(B) Bio-gas
(C) Phosgene gas
(D) Mustard gas

Q4. Which three elements are needed for the healthy growth of plants?

(A) N , S , P
(B) N , Ca , P
(C) N , P , K
(D) N , K , C

Q5. The reaction between fat and NaOH is called

(A) Esterification
(B) Hydrogenolysis
(C) Fermentation
(D) Saponification

Q6. Which compound is more soluble in water?

(A) C_2H_5OH
(B) C_6H_5OH
(C) CH_3COCH_3
(D) n–hexanol

Q7. Which one of the following ions has maximum value of heat of hydration?

(A) Na^+
(B) Cs^{2+}
(C) Ba^{2+}
(D) Mg^{2+}

Q8. Hydrogen bond is the strongest between the molecules of

(A) HF
(B) HCl
(C) HBr
(D) HI

Q9. Tincal is a mineral of

(A) Al
(B) B
(C) Si
(D) C

Q10. The amount of Ozone in the atmosphere is expressed in units

(A) KJ
(B) KJ/mole
(C) DU
(D) N

Q11. The solution of which acid is used for seasoning of food?

(A) Formic acid
(B) Acetic acid
(C) Benzoic acid
(D) Butanoic acid

Q12. The state of hybridization of Carbon atom in Methane is

(A) sp^3
(B) sp^2
(C) sp
(D) dsp^2

Q13. Peroxyacetyl nitrate (PAN) is an irritant to human beings and it affects

(A) Eyes
(B) Ears
(C) Stomach
(D) Nose

Q14. Choose the gas which is obtained by the reaction of Ethyl alcohol with conc. H_2SO_4

(A) CO
(B) CO_2
(C) C_2H_2
(D) C_2H_4

Q15. Select the two normal elements present in sixth period

(A) K , Ca
(B) Rb , Sr
(C) Cs , Ba
(D) La , Hf

Q16. Cannizzaro's reaction is not given by

(A) Formaldehyde
(B) Acetaldehyde
(C) Benzaldehyde
(D) Trimethylacetaldehyde

Q17. When water (H – OH) is made to react with Ethyl Magnesium Bromide, the product formed is

(A) $CH_2 = CH_2$
(B) $HC \equiv CH$
(C) $CH_3 – CH_3$
(D) CH_4

Chemistry (Intermediate Part-II) Paper : II (Subjective Type)

Time Allowed: 2.40 hours | **Maximum Marks:** 68

Note: Section I is compulsory. Attempt any THREE (3) questions from Section II.

Q2. (Marks (2 x 8 = 16)) Write short answers to any EIGHT questions.

i. Why diamond is non-conductor and graphite is fairly a good conductor?

ii. Write the names of families in periodic table.

iii. Why 'd' and 'f' block elements are called transition elements?

iv. How $KMnO_4$ and $K_2Cr_2O_7$ react with H_2S?

v. Write the chemistry of setting of cement in between 1 – 7 days.

vi. Justify that BeO is amphoteric in nature.

vii. What will happen when Magnesium reacts with (i) H_2 (ii) N_2

viii. Starting from primary alkyl halide, prepare ethyl alcohol.

ix. What is degree of polymerization?

x. What are the oligosaccharides? Give one example.

xi. What do you mean by denaturation of protein?

xii. How does Grignard reagent react with CO_2?

Q3. (Marks (2 x 8 = 16)) Write short answers to any EIGHT questions.

i. Write down the structures of N_2O_3 and NO_2.

ii. Why the elements of group VI A other than Oxygen show more than two oxidation states?

iii. Write down any four uses of bleaching powder.

iv. Define disproportionation reaction with an example.

v. Define sp hybridization with an example.

vi. How wood is converted into coal? Give its equation too.

vii. What is Wolf-Kishner's reduction? Give reaction.

viii. How is PCl_3 produced from $SOCl_2$?

ix. How chloroprene is prepared from acetylene? Give reaction, also.

x. How aeration is used for purification of H_2O?

xi. What is acid rain? How does it affect on aquatic life?

xii. How smog is produced? Write down its conditions of formation.

Q4. (Marks (2 x 6 = 12)) Write short answers to any SIX questions.

i. What is the action of an aqueous solution of borax on litmus?

ii. Give any two uses of boric acid.

iii. How and under what conditions does Aluminium react with the following : (a) Halogens (b) Alkalies

iv. What is meant by the terms : (a) Nitration (b) Halogenation

v. Why Ethanol has higher boiling point than diethyl ether?

vi. How will you distinguish between methanol and ethanol?

vii. Give the mechanism of addition of HCN to acetone.

viii. How to prepare acetic acid from ethyne?

ix. What are zwitter ions?

> Note: SECTION – II

Q5.

(a) [4] What are hydrides? Explain types of hydrides with their properties.

(b) [4] What is the role of lime in agriculture and ceramic industries?

Q6*.

(a) [4] Write down any eight applications of noble gases in daily life.

(b) [4] What is significance of potassium fertilizers in plant growth? Explain the manufacturing of KNO_3 on industrial scale.

Q7.

(a) [4] Explain any four types of structural isomerism by giving one example of each.

(b) [4] What are alkyl halides? How are the alkyl halides prepared from alcohol by three different reactions?

Q8.

(a) [4] Write a note on photochemical halogenation of methane.

(b) [4] Describe the Reduction reactions of Carbonyl Compounds with following reagents: i. $NaBH_4/H_3O^+$ ii. Ni/H_2 (Give two reactions for each)

Q9.

(a) [1+3=4] What is Kekule's Structure of Benzene? How did he support his theory?

(b) [4] How will you prepare phenol from the following? i. Chlorobenzene ii. Sodium salt of Benzene Sulphonic Acid